

CallMeMaybe – Operator Efficiency Analysis

Subtitle:

Identifying inefficient operators using call activity data

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Tableau

Link:https://public.tableau.com/app/profile/cristian.julian.rodriguez.valbuena/viz/CallMeMaybeOperatorPerformanceDashboard_17894336740170/Dashboard1

Business Problem and Objective

Business Problem

- CallMeMaybe wants to help supervisors identify inefficient operators.
- Inefficiency may be reflected in:
 - High missed incoming-call rates.
 - Long waiting times for incoming calls.
 - Low outgoing-call activity when outbound work is expected.

Objective

- Define measurable inefficiency criteria.
- Identify operators who consistently underperform.
- Support supervisors with actionable recommendations for monitoring and improvement.

Data and Methodology

Data

- 49,002 cleaned call records
- 307 client accounts
- 1,092 identified operators
- Analysis period: August 2 to November 28, 2019
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Methodology

- Clean and validate the telecom data.
- Analyze call direction, missed calls, waiting times, and operator workload.
- Build operator-level performance metrics.
- Define inefficiency thresholds using data percentiles.
- Classify inefficient and high-priority operators.
- Validate the differences with statistical hypothesis tests.

Key EDA Findings

- Outgoing calls represent about **75%** of total call volume.
- External calls account for approximately **98%** of activity.
- Around **52.7% of incoming calls were missed**.
- Incoming-call waiting times are strongly right-skewed, with clear high-value outliers.
- Outgoing-call activity also varies widely across operators, with a small number handling very large volumes.

Business takeaway

- Missed incoming calls and waiting time are meaningful operational issues.
- Operator activity is highly heterogeneous, so performance must be evaluated with role-relevant metrics rather than a single universal rule.

Operator Inefficiency Criteria

An operator is flagged as potentially inefficient if at least one of the following conditions is met:

- High missed incoming-call rate: above 5.08%
- Long average waiting time: above 21.13 seconds
- Low outgoing activity: fewer than 11 outgoing calls, considering only operators with outbound activity

Incoming-call thresholds were calculated using operators with at least 5 incoming calls to reduce instability caused by very low call volumes.

Priority level

- Potentially inefficient: meets at least 1 criterion
- High priority: meets 2 or more criteria

Important note

- Operators with zero outgoing calls are not automatically considered inefficient, since some may primarily handle incoming calls.

Potentially Inefficient Operators Identified

- 1,092 operators were evaluated.
- 363 operators were classified as potentially inefficient.
- 729 operators did not meet any inefficiency criterion.
- 41 operators were classified as high priority, meeting at least two criteria.

Inefficiency score distribution

- Score 0: 729 operators
- Score 1: 322 operators
- Score 2: 39 operators
- Score 3: 2 operators

Business takeaway

- Most potentially inefficient operators fail on only one criterion.
- The 41 high-priority operators should be reviewed first because they show multiple performance issues at the same time.

Efficient vs Potentially Inefficient Operators

Efficient operators

- Missed incoming-call rate: 1.01%
- Average waiting time: 14.08 seconds
- Average outgoing calls: 790.59

Potentially inefficient operators

- Missed incoming-call rate: 3.16%
- Average waiting time: 23.89 seconds
- Average outgoing calls: 88.17

Key finding

- Potentially inefficient operators show a substantially higher missed-call rate.
- Their average waiting time is notably longer.
- They also handle far fewer outgoing calls on average.

Statistical Hypothesis Testing

To assess whether the differences between efficient and potentially inefficient operators were statistically significant, two one-sided Mann–Whitney U tests were performed using operators with at least 5 incoming calls.

Hypothesis 1 — Missed incoming-call rate

- Efficient operators: 341
- Potentially inefficient operators: 207
- Mann–Whitney U = 42,676.5
- p-value $\approx 2.38 \times 10^{-6}$
- Result: Reject H_0
- Potentially inefficient operators have a significantly higher missed incoming-call rate.

Hypothesis 2 — Average waiting time

- Efficient operators: 341
- Potentially inefficient operators: 207
- Mann–Whitney U = 59,888.5
- p-value $\approx 6.14 \times 10^{-43}$
- Result: Reject H_0
- Potentially inefficient operators have significantly longer average waiting times.

Key takeaway

- The statistical tests confirm strong differences between the two groups for the main incoming-call performance indicators.
- Because the classification framework is partly based on these same performance metrics, the tests should be interpreted as confirming strong statistical separation between the groups rather than as independent causal validation.

Recommendations and Final Conclusion

Recommendations

- Monitor missed-call rate and average waiting time together rather than relying on a single metric.
- Prioritize the 41 high-priority operators for supervisory review.
- Review workload distribution and call-routing processes for operators with unusually long waiting times.
- Provide targeted training to operators with elevated missed incoming-call rates.
- Investigate low outgoing activity only when outbound work is part of the operator's role.
- Use the inefficiency score as an ongoing screening and monitoring framework rather than as an automatic performance judgment.

Final Conclusion

- 363 of 1,092 operators were identified as potentially inefficient.
- 41 operators were classified as high priority because they met at least two inefficiency criteria.
- Potentially inefficient operators showed higher missed-call rates, longer average waiting times, and lower outgoing-call activity.
- Statistical testing confirmed strong differences between the groups for the main incoming-call performance indicators.
- The proposed framework can help supervisors focus monitoring and improvement efforts on operators with the greatest operational risk.